Crypto Lab 1 - Shishir Vuppala

Challenge 3: PRGone Wrong

flag: `cs409{4lway5_7h1nk_1in34r1y}`

Solution

The generator follows the recurrence

$$S_{i+1} = AS_i + C \pmod{M},$$

The first state S_0 is the flag and so we have

$$S_1 = AS_0 + C \pmod{M},$$

$$(S_1 - C) = AS_0 \pmod{M},$$

$$\text{flag} = S_0 = A^{-1}(S_1 - C) \pmod{M},$$

Note: $\gcd(A, M) = 1$, and so there exists an inverse of A modulo M. To calculate this inverse, we need the value of $\phi(M)$, which is $M/2$ since M is a power of 2. Then $A^{\phi(M)-1}$ is A^{-1} modulo M. Python does this automatically using `pow(A,-1,M)`.

Since we don't know the exact value of S_1 because last 8 bits were removed. we shift it to the left 8 bits and check all 256 possibilities and see which of them decode to a ASCII string which has 'cs409{' at the starting.

Code

The script used to solve this challenge is written in `decrypt.py`